Development of the St Mary's University Urban Mobility Hub

Regnum: **2501608**

Module code: **CPS4006 Web Design and Development**

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Introduction

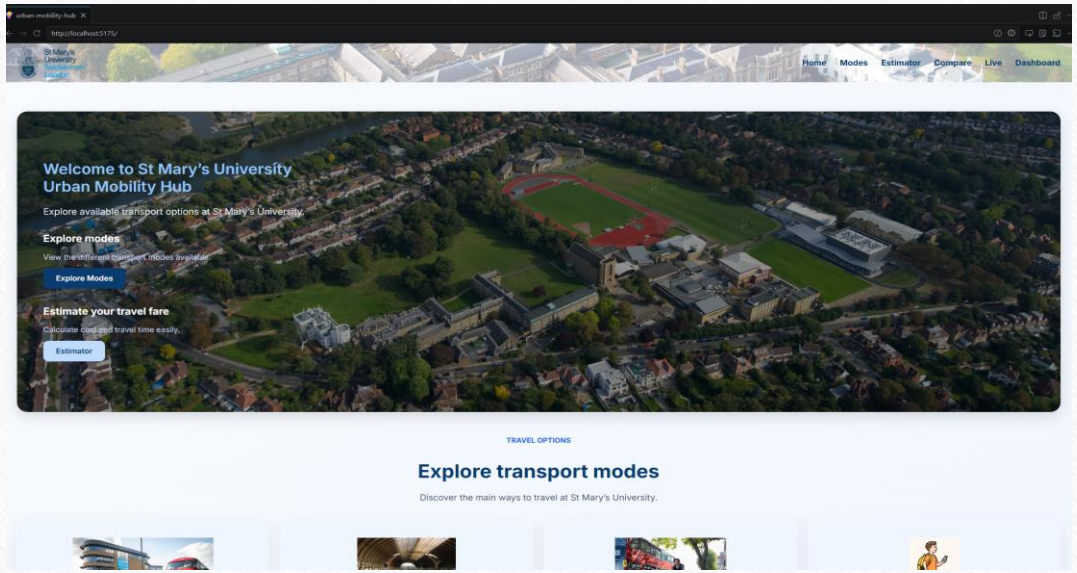
The Urban Mobility Hub is a React based web application designed to help students and visitors plan transport journeys around London using live TfL data.

- • Developed using React and Vite
  • Integrated with the TfL API
  • Provides live transport updates
  • Includes estimator helps to plan journey and comparison tools
  • Responsive mobile friendly design

Introduction

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Project Objectives

- Build responsive React application
- Integrate TfL API
- Compare travel options
- Implement local storage
- Support desktop and mobile devices

Technologies Used

- React
- Vite
- JavaScript
- CSS3
- React Router
- TfL API
- GitHub

System Architecture

- Component based architecture
- Reusable components
- React Router navigation
- useState and useEffect hooks

```
3  import { BrowserRouter } from "react-router-dom"
4  import App from "./App"
5  import "./index.css"
6
7  ReactDOM.createRoot(document.getElementById("root")).render(
8    <BrowserRouter>
9      <App />
10   </BrowserRouter>
11 )
```

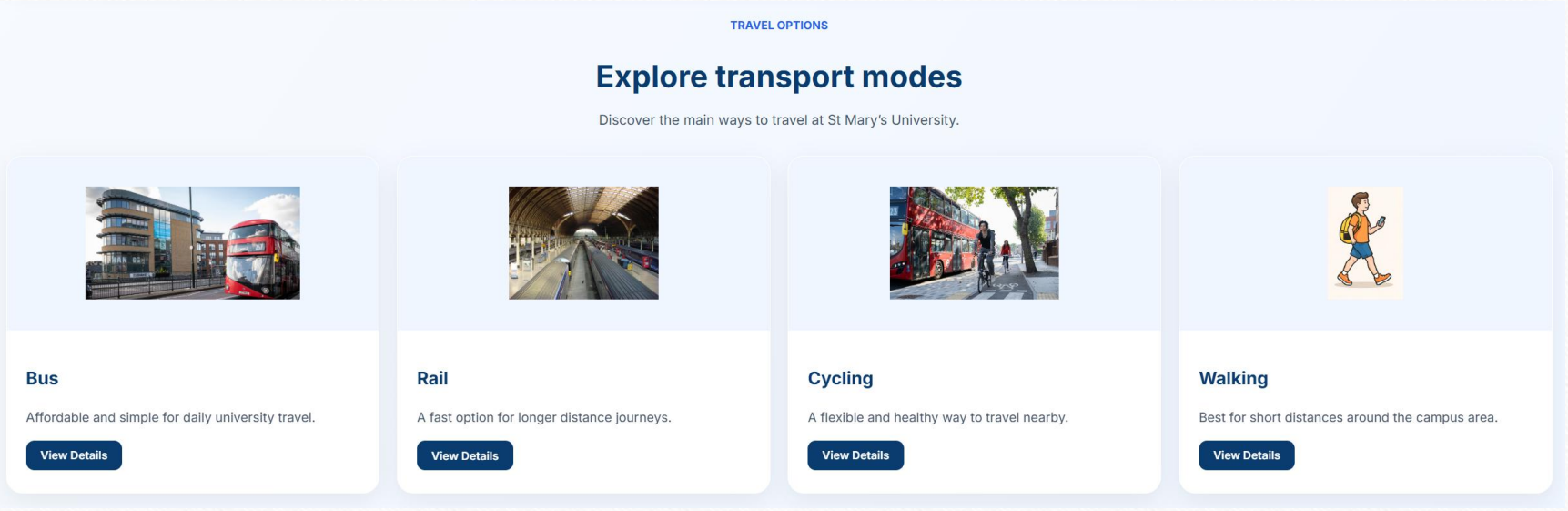
```
24       <Routes>
25         <Route path="/" element={<Home />} />
26         <Route path="/modes" element={<Modes />} />
27         <Route path="/estimator" element={<Estimator />
28         <Route path="/compare" element={<Compare />} />
29         <Route path="/dashboard" element={<Dashboard />
30         <Route path="/live" element={<LiveTransport />
31       </Routes>
32     </div>
```

Transport Modes Feature

- Dynamic rendering using map()
- Transport advantages and disadvantages
- Reusable card layout

Transport Modes Feature

- Dynamic rendering using map()
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Transport overview page : Bus

TRANSPORT MODES

Explore Transport Options

Learn about different ways to travel and choose the best option.



Bus

Affordable daily travel for medium distance journeys.

Pros

- Affordable
- Useful for medium trips
- Easy access

Cons

- Traffic delays
- Can be crowded

Fare depends on travel distance.

Transport overview page : Rail



Rail

Fast transport for long journeys.

Pros

- Fast
- Reliable
- Comfortable

Cons

- Expensive
- Limited routes

Best option for long distance travel.

Transport overview page : Cycling



Cycling

Eco friendly and flexible travel option.

Pros

- Low cost
- Healthy
- Environment friendly

Cons

- Weather issues
- Takes longer

Personal bicycle is free. Shared bikes may cost a small fee.

Transport overview page : Walking



Walking

Free and healthy for short distances.

Pros

- Free
- Healthy
- Simple

Cons

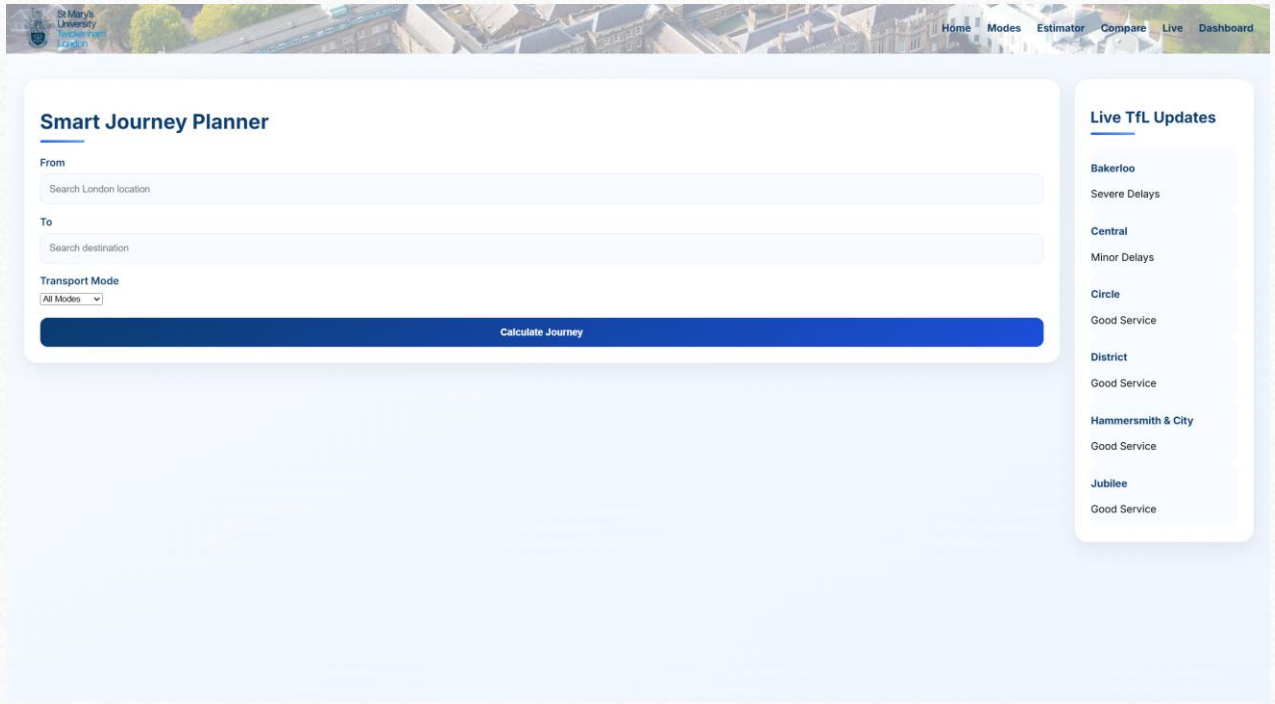
- Slow
- Weather dependent

Best for nearby locations.

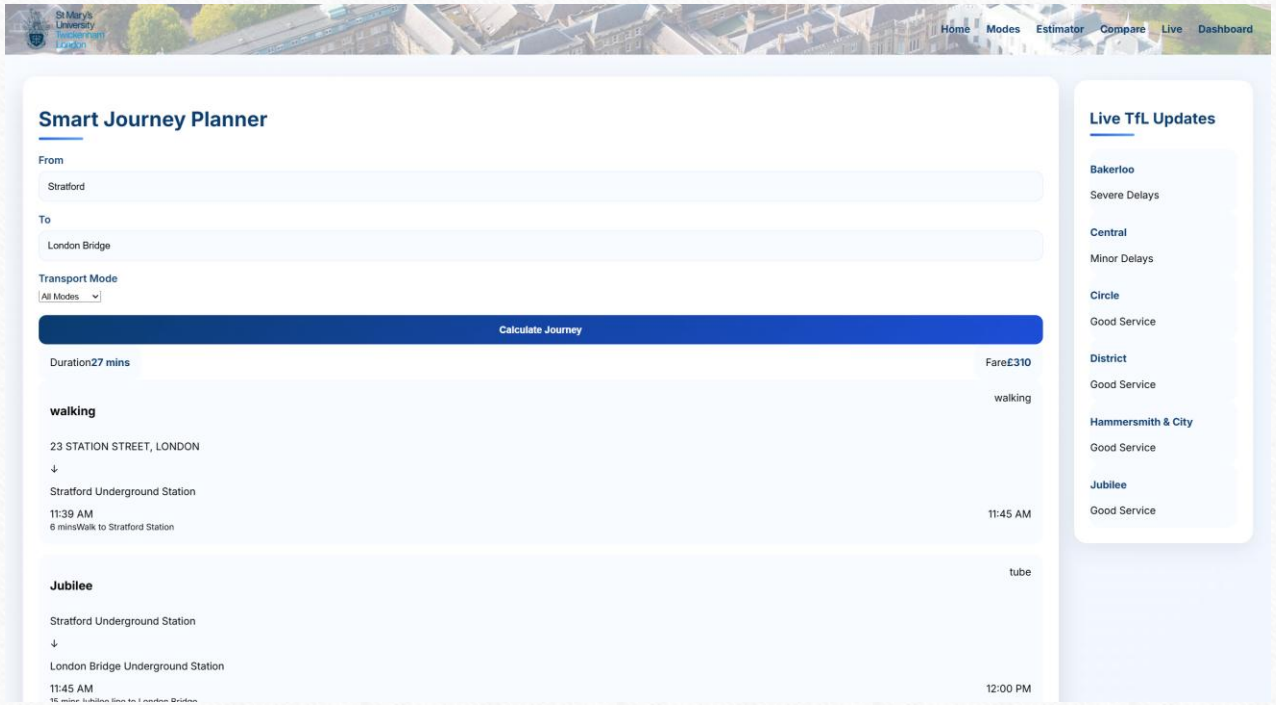
Estimator

- TfL Journey Planner API
- Autocomplete search
- Real time route calculations
- Fare and duration estimates

Empty planner



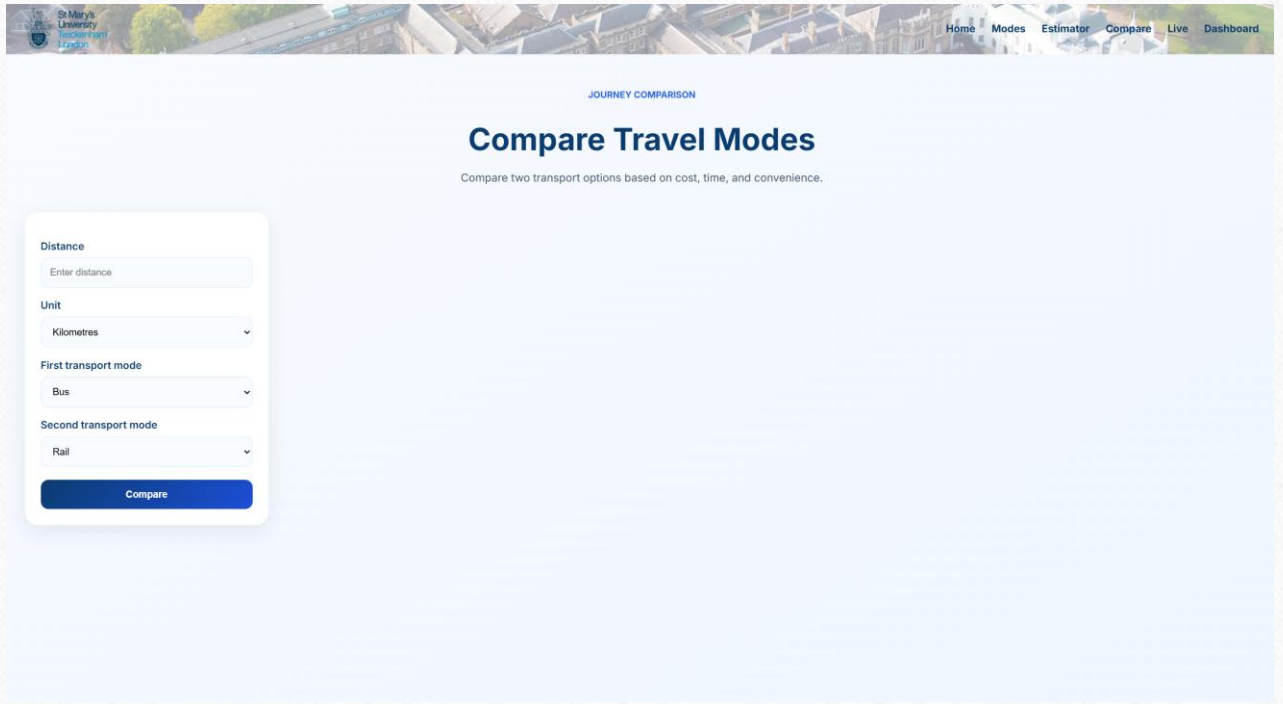
Journey results



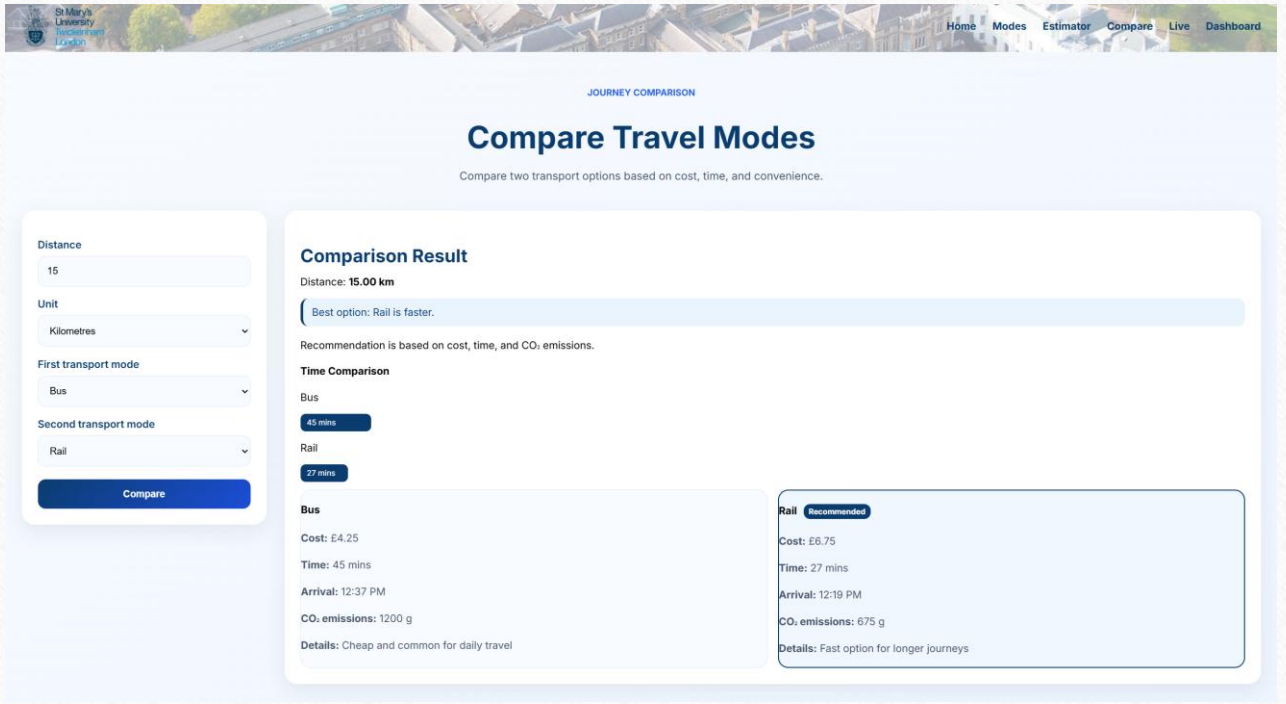
Transport Comparison

- Compare cost
- Compare travel time
- CO_2 emissions
- Recommendation system

Compare interface



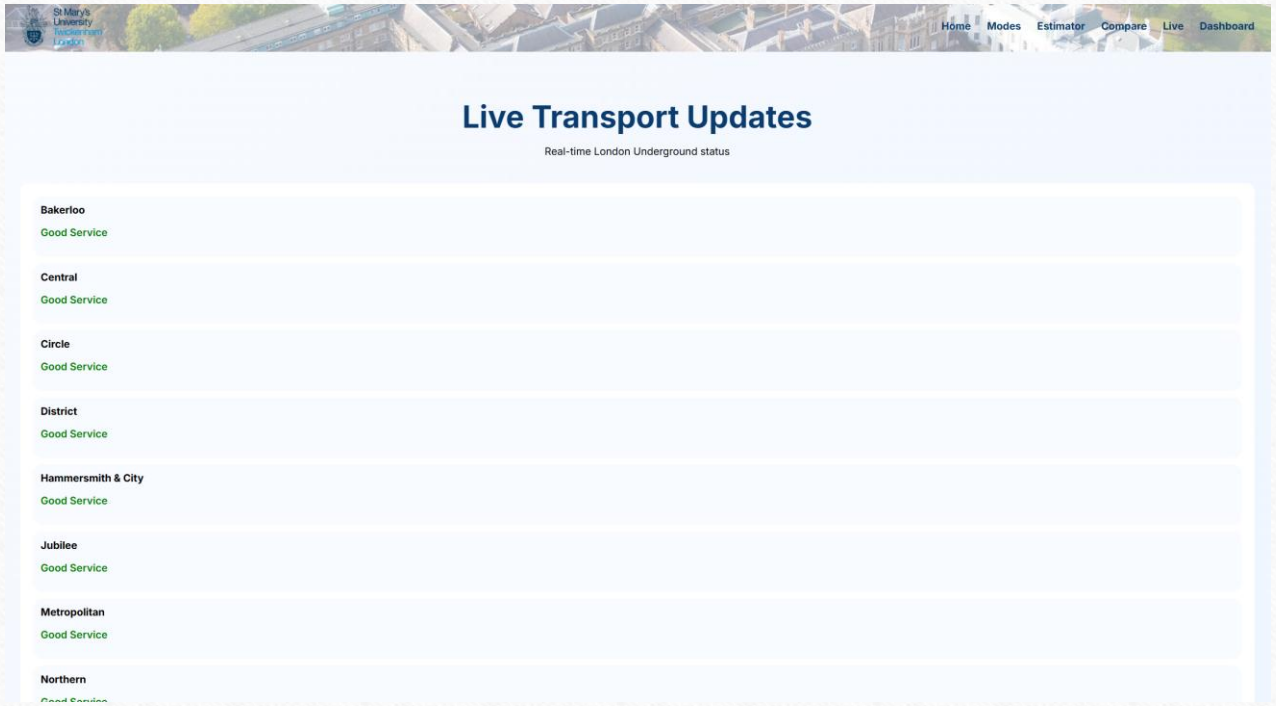
Comparison results



Live Transport Updates

- Real time TfL underground status
- Automatic refresh every 10 seconds
- Delay monitoring

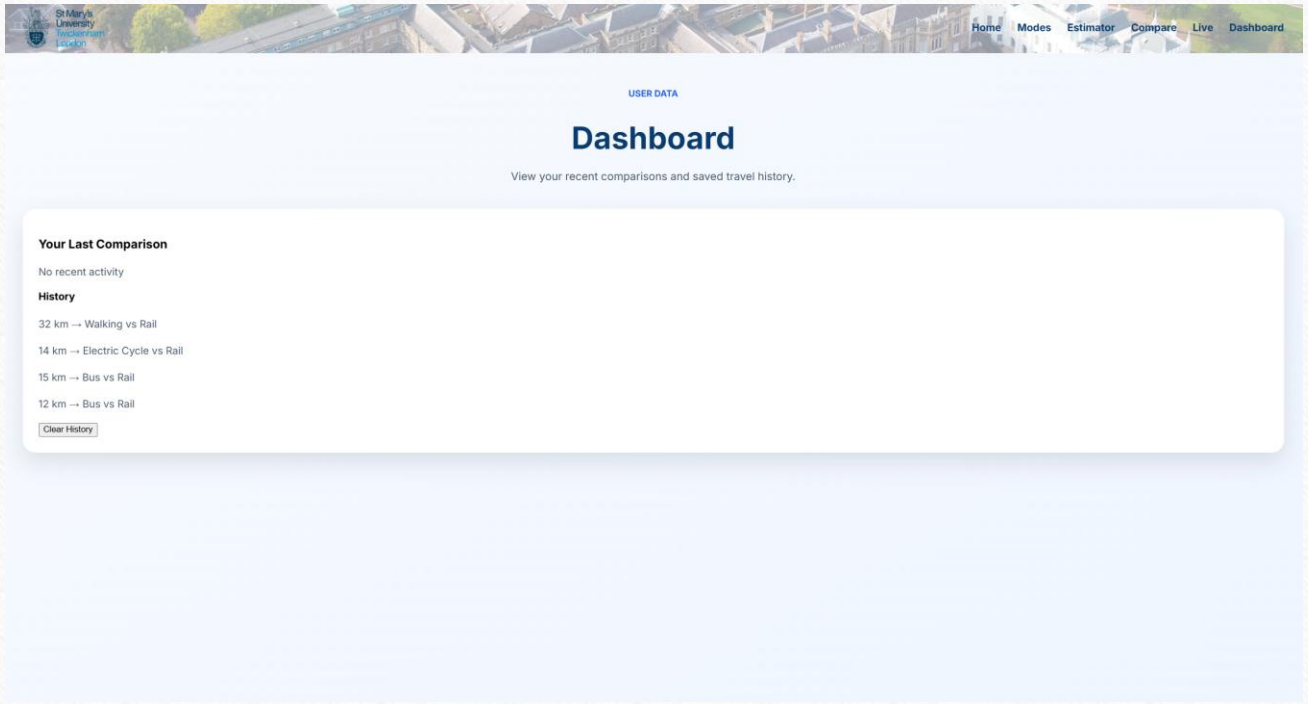
Live updates page



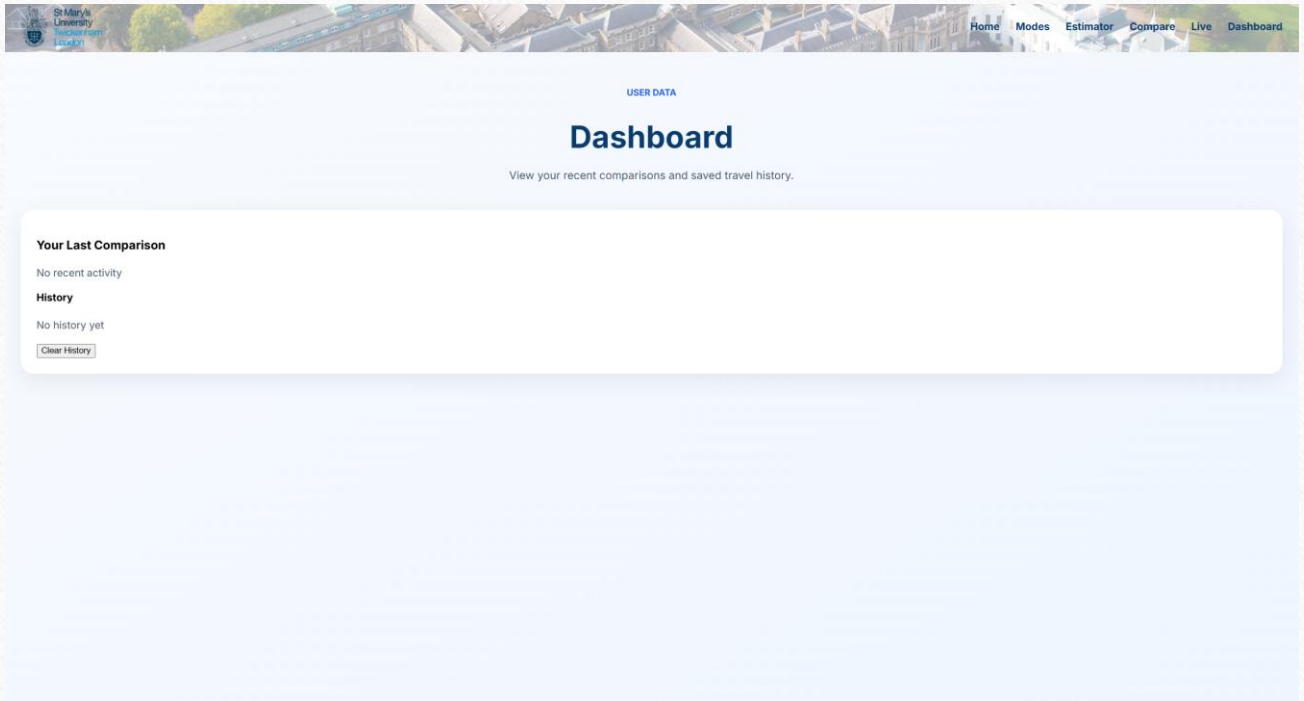
Dashboard And Local Storage

- Browser local storage
- Saved comparison history
- Clear history function

Dashboard with history



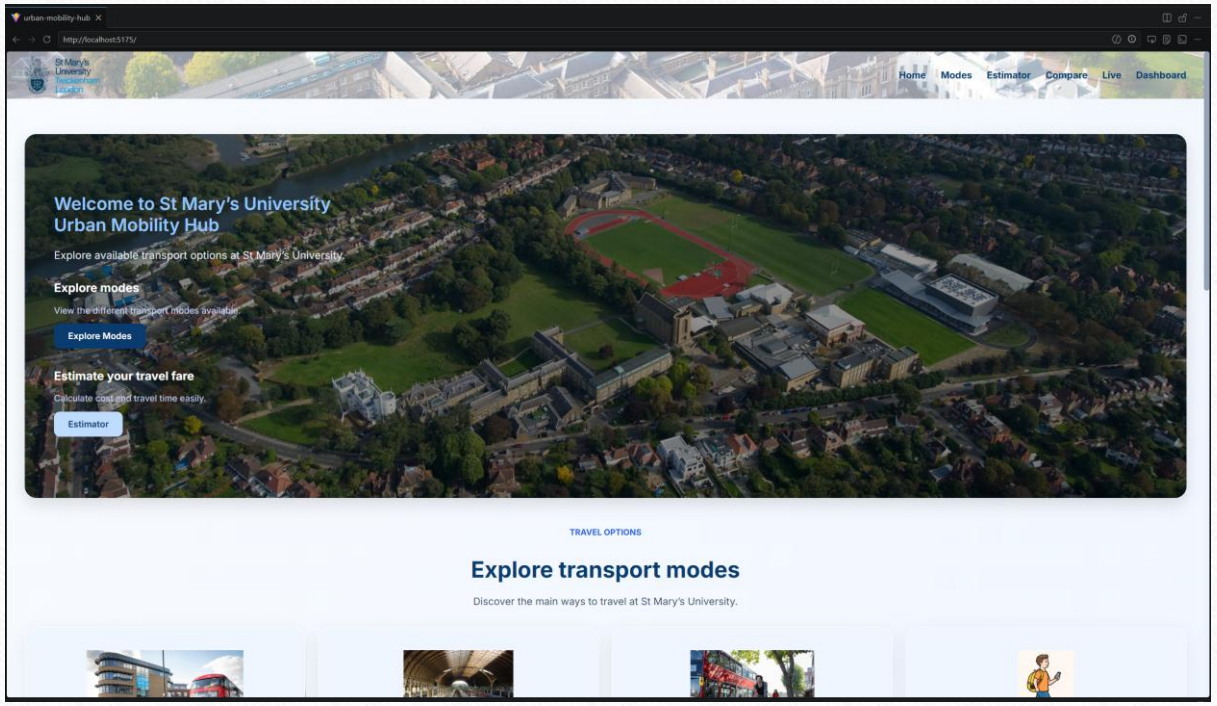
Cleared history



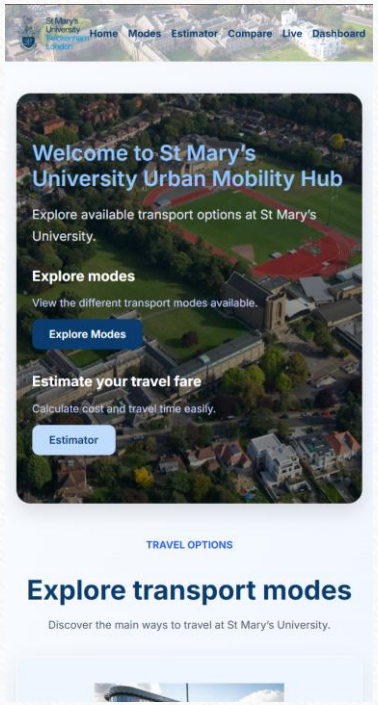
Responsive Design

- CSS media queries
- Responsive layouts
- Mobile optimisation

Desktop view



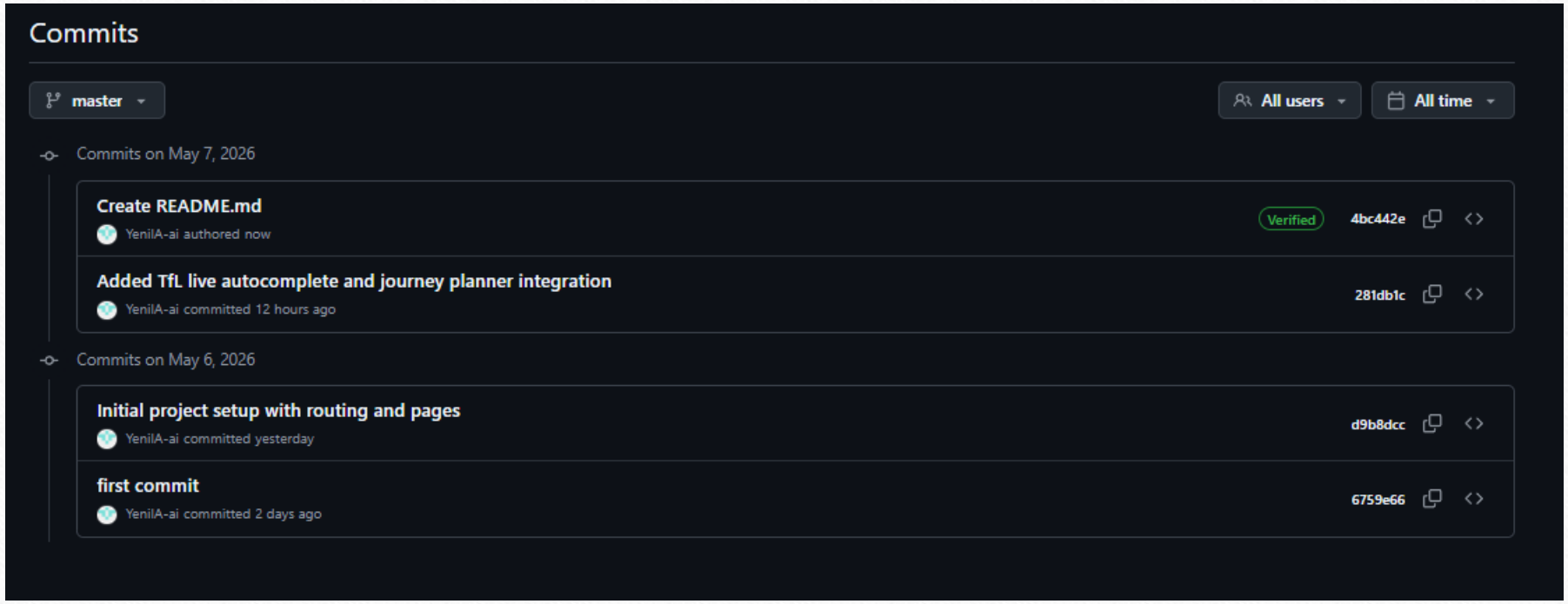
Mobile view


Home
Modes
Estimator
Compare
Live
Dashboard
Welcome to St Mary's University Urban Mobility Hub
Explore available transport options at St Mary's University.
Explore modes
View the different transport modes available.
Explore Modes
Estimate your travel fare
Calculate cost and travel time easily.
Estimator
TRAVEL OPTIONS
Explore transport modes
Discover the main ways to travel at St Mary's University.

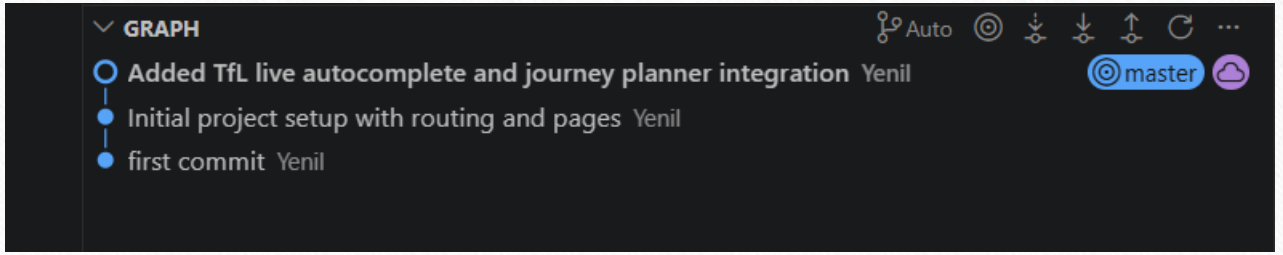
Github Repository And Version Control

- Source code management
- Commit tracking
- Project backup

GitHub repository



GitHub commit history showing project development progress



Challenges And Solutions

Challenge:
- TfL API integration
- Handling asynchronous data
- Responsive layout adjustments

Solutions:
- React hooks
- Conditional rendering
- CSS media queries

Ethical And Professional Considerations

- Accurate transport information
- User friendly design
- Accessibility
- Clear navigation

Testing And Evaluation

- Navigation testing
- API testing
- Responsive testing
- Local Storage testing
- TfL API response Testing

Future Improvements

- Google Maps integration
- User authentication
- Dark mode
- Additional transport providers (e.g. Maps)

Conclusion

- React frontend application
- Live transport integration
- Responsive user interface
- Successful project objectives

THANK YOU